A short proof of a lower bound for Turán numbers

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Abstract

Let F be a strictly balanced r-uniform hypergraph with $e > 2$ edges and r-density m. We give a new short proof of the fact that the Turán number $\mathrm{ex}(n, F)$ is greater than $c\,n^{r-1/m}(\log n)^{1/(e-1)}$ where c depends only on F. The previous proof of this for $r = 2$ by Bohman and Keevash and for $r \geq 3$ by Bennett and Bohman used a random greedy process and its analysis using the differential equations method. Our proof uses elementary probabilistic arguments together with a (nontrivial) classical result about independent sets in hypergraphs.

1 Introduction

Given an r-uniform hypergraph F (henceforth r-graph), let m_F be the maximum, over all subgraphs F' of F with at least $r+1$ vertices, of $(e'-1)/(v'-r)$, where v' and e' are the number of vertices and edges of F'. Say that F is strictly balanced if F has at least $r+1$ vertices and m_F is uniquely achieved by $F' = F$. Write $\mathrm{ex}(n, F)$ for the maximum number of edges in an r-graph that contains no copy of the r-graph F.

Brown-Erdős-Sós [4] proved that for any r-graph F with $v > r$ vertices and $e > 1$ edges, there exists c such that $\mathrm{ex}(n, F) > c\,n^{r-(v-r)/(e-1)}$. The first improvement to this general bound was achieved by Wolfowitz [7] who proved that if F is a strictly balanced regular graph, then $\mathrm{ex}(n, F) > cn^{2-1/m_F}(\log\log n)^{1/(e-1)}$. Bohman and Keevash [3] improved the log factor and removed the regularity requirement and then Bennett and Bohman [2] proved the the same result for r-graphs.

Theorem 1. (Bohman-Keevash [3], Bennett-Bohman [2]) *Let $r \geq 2$ and F be a strictly balanced r-graph with $v > r$ vertices and $e > 2$ edges. Then there exists c such that* $\mathrm{ex}(n, F) > c\,n^{r-1/m_F}(\log n)^{1/(e-1)}$.

The proofs in [3] and [2] are highly nontrivial, and employ the differential equations method to analyze the random greedy independent set process on hypergraphs. Very recently, Ferber-Mckinley-Samotij [6] conjectured that the bound in Theorem 1 is not sharp in order of magnitude if F is a graph containing a cycle. Still, Theorem 1 gives the best known result for a typical degenerate Turán problem, e.g. for $\mathrm{ex}(n, K_{t,t})$ when $t \geq 5$.

In this note, we prove Theorem 1 by a very short probabilistic argument and the following theorem of Duke-Lefmann-Rödl [5] that extends the celebrated result of Ajtai-Komlós-Pintz-Spencer-Szemerédi [1]. A hypergraph is linear if every two vertices lie in at most one edge.

Theorem 2. (Duke-Lefmann-Rödl [5]) *For all $k \geq 3$ there exists $c > 0$ such that every linear k-graph with n vertices and average degree at most d has an independent set of size at least $c\,n(\log d/d)^{1/(k-1)}$.*

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Our proof of Theorem 1 is very simple extension of that of Brown-Erdős-Sós [4]. Their method considers the random r-graph $G = G_r(n,p)$ where p is chosen so that the number of copies of F is of the same order as the number of edges, and the final graph G' is obtained by deleting an edge for each copy of F. In our proof, we increase p slightly so that we have many more edges and even more copies of F, but we delete some edges so that these copies of F are distributed fairly evenly. In the final step, we pick a subset of edges that avoids all copies of F, gaining a log factor by using Theorem 2. In what follows, all asymptotic notation is with respect to $n \to \infty$.

Proof of Theorem 1. For $r < i \le v$, let $\mathcal{F}_i$ be the set of r-graphs that comprise two copies of F that share exactly i vertices; in particular, the edge-sets of the two copies of F forming any $F_i \in \mathcal{F}_i$ cannot be identical. Each $F_i \in \mathcal{F}_i$ has $2v - i$ vertices and at least $e_i = 2e - t_i$ edges where t_i is the maximum number of edges that an i-vertex subgraph of F can have. Suppose that $i < v$. Since F is strictly balanced, $(t_i - 1)/(i - r) < m := m_F$ and we may define $\alpha_i < 1$ by $t_i = m(i - r) + \alpha_i$. Now choose any ϵ that satisfies

$$0 < \epsilon < \min_{r<i<v} \left\{ \frac{1 - \alpha_i}{m(2vm - mr - mi + 1 - \alpha_i)}, \frac{1}{m} - \frac{v - r}{e} \right\} < \frac{1}{m}. \tag{1}$$

Let $G = G_r(n,p)$ be the random r-graph on $[n]$ with edge probability $p = n^{\epsilon - 1/m} \in (0,1)$. Let $X = |E(G)|$, Y be the number of copies of F in G, and Z_i be the number of copies of any $F_i \in \mathcal{F}_i$ in G. Note that Z_v counts v-vertex subgraphs that contain two copies of F, and therefore contain at least $e + 1$ edges. Clearly

$$\mathbb{E}[X] = p\binom{n}{r}, \qquad \mathbb{E}[Y] = O(n^v p^e), \qquad \mathbb{E}[Z_i] = O(n^{2v-i} p^{2e - m(i-r) - \alpha_i}), \qquad \mathbb{E}[Z_v] = O(n^v p^{e+1}).$$

Let $Z = \sum_{i=r+1}^{v} Z_i$. Since $e > 1$ and F is strictly balanced, $\epsilon - 1/m + r > 0$ and so $pn^r \to \infty$. Therefore, by Chebyshev's inequality (for X), and Markov's inequality (for Y, Z),

$$\mathbb{P}[X < (0.9)\,\mathbb{E}[X]] < 1/3, \qquad \mathbb{P}[Y > 3\,\mathbb{E}[Y]] < 1/3, \qquad \mathbb{P}[Z > 3\,\mathbb{E}[Z]] < 1/3.$$

Hence there exists an n vertex r-graph G for which $X \ge (0.9)\,\mathbb{E}[X], Y \le 3\,\mathbb{E}[Y]$ and $Z \le 3\,\mathbb{E}[Z]$. Using (1) and $e = m(v - r) + 1$, yields $2v + (\epsilon - 1/m)(2e + mr - \alpha_i) - \epsilon mi < \epsilon - 1/m + r$ and consequently

$$\mathbb{E}[Z_i] = O(n^{2v} p^{2e + mr - \alpha_i} (p^m n)^{-i}) = O(n^{2v} p^{2e + mr - \alpha_i} n^{-\epsilon mi}) = o(pn^r) = o(\mathbb{E}[X])$$

for $i < v$. Also, $\epsilon < 1/m - (v - r)/e$ yields $\mathbb{E}[Z_v] = o(\mathbb{E}[X])$, so $\mathbb{E}[Z] = o(\mathbb{E}[X])$. The r-graph G therefore has $X - Z > (0.8)\,\mathbb{E}[X] > cpn^r$ and $Y \le 3\,\mathbb{E}[Y] < Cn^v p^e$ for positive constants c, C. Removing an edge from each copy of F_i counted by Z, we obtain an n vertex r-graph $G' \subset G$ with:

1) $|E(G')| > cpn^r$

2) the number of copies of F in G' is at most $Cn^v p^e$

3) $F_i \not\subset G'$ for all $F_i \in \mathcal{F}_i$ and $r < i \le v$.

Let $H = (V, E)$ be the e-graph where $V = E(G')$ and E consists of the edge sets of the copies of F in G'. If $v, v' \in V = E(G')$ and there are two copies of F in G' containing the edges v, v', then we obtain a copy of some $F_i \in \mathcal{F}_i$ in G' where $r < i \le v$, as the intersection of these two copies contains $v \cup v'$ which has size at least $r + 1$. Since $F_i \not\subset G'$, we conclude that H is linear. The average degree of H is

$d := d(H) = O(|E|/|V|) = O(n^{v-r}p^{e-1})$. Since $n^{v-r}p^{e-1} = n^{\epsilon(e-1)}$, Theorem 2 implies that H has an independent set I of size at least

$$c_1|V|\left(\frac{\log d}{d}\right)^{1/(e-1)} > c_2\frac{pn^r}{(n^{v-r}p^{e-1})^{1/(e-1)}}(\log(n^{v-r}p^{e-1}))^{1/(e-1)} > c_3\, n^{r-1/m}(\log n)^{1/(e-1)}.$$

The subgraph of edges in G' corresponding to the vertices in I contains no copy of F. $\square$

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